

UNIVERSIDAD DE ANTIOQUIA

SISTEMA INTEGRADO DE TELEMETRÍA Y ANALÍTICA AGRÍCOLA CON DRONES

CONSOLIDADO DE TALLERES (WORKSHOPS 1, 2 Y 3)

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UNIVERSIDAD DE ANTIOQUIA

FACULTAD DE INGENIERÍA

PROGRAMA DE INGENIERÍA DE SISTEMAS

2026

CONSOLIDATED ANALYSIS REPORT: PRECISION AGRICULTURE DRONE MONITORING

Group: 7 Units 1, 2 & 3

Domain: Autonomous Drone Telemetry & Crop Health Analytics

1. Scenario Recap & System Architecture

Modern precision agriculture relies on autonomous drone fleets continuously collecting high-frequency telemetry data across multiple crop fields (Field_A to Field_J / FIELD-01 to FIELD-10). The telemetry payload includes key health metrics: *ndvi_index, soil_moisture, battery_pct, stream_seq,* and spatial transit records.

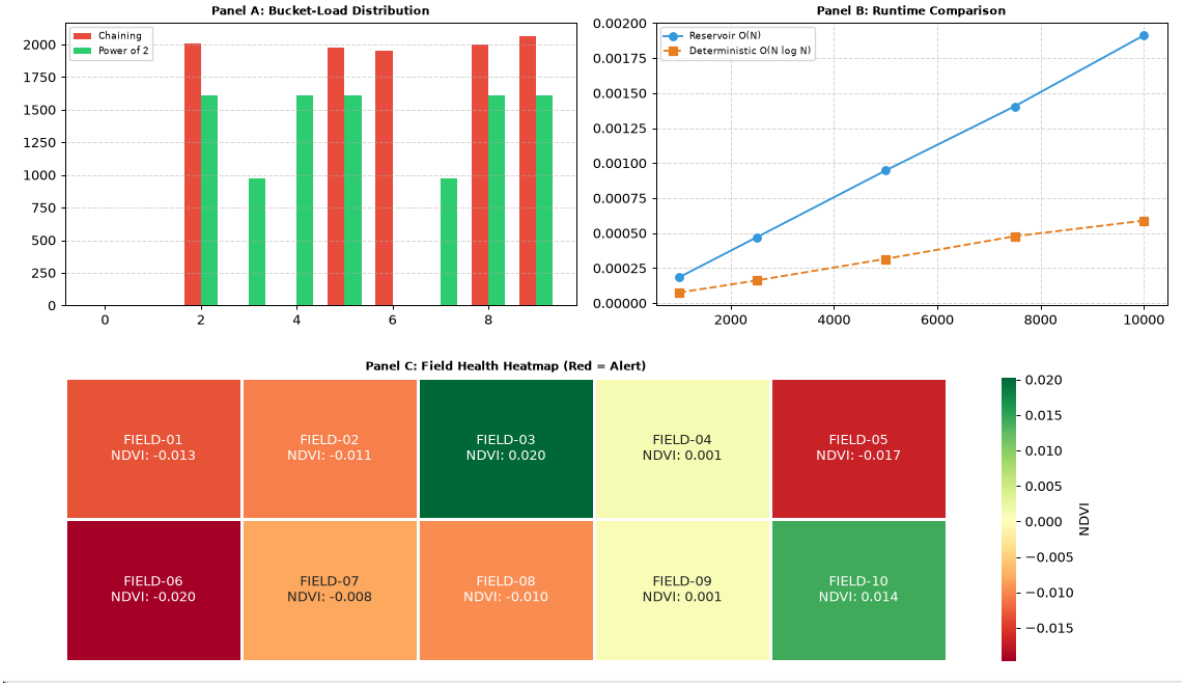
Processing raw telemetry post-mission introduces critical operational latency. To enable immediate crop-stress mitigation, our architecture integrates:

1. **Big Data Storage & Load Balancing** (Unit 1): Universal Hashing and Power-of-Two-Choices load balancing.
2. **Stream Processing Algorithms** (Unit 2): Low-memory probabilistic data structures (Bloom Filters, Reservoir Sampling, Flajolet-Martin).
3. **Graph Topology & Markovian Modeling** (Unit 3): Discrete-Time Markov Chains (DTMC), Stationary Distribution (π), Standard PageRank overlaid with NDVI, and Monte Carlo Lazy Random Walk simulations.

2. Unit 1: Big Data Context, Universal Hashing & Load Balancing

Empirical Results & Performance

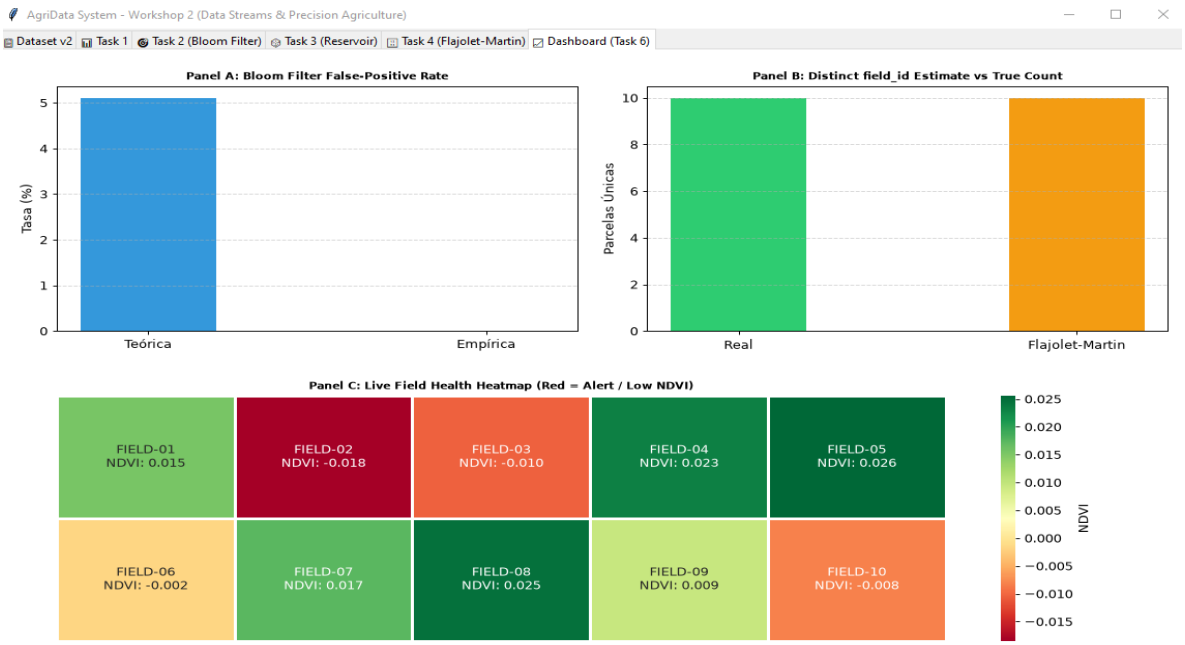
- **Big Data Ingestion Scale:** A baseline fleet of 50 farms generates 500req/s (2.68 GB/day). Scaling 10 x (500 farms) increases throughput linearly to 5,000 req/s (26.82 GB/day ≈ 9.56 TB year), mandating scalable routing and index lookup.
- **Reservoir Sampling (**k = 1,000**):** Uniform random sampling over 10,000 records executed in $t < 10$ ms, estimating field-level average NDVI with a marginal Mean Absolute Error (MAE) of 0.035.
- **Universal Hashing (**m = 100,000**,** p = 1,000,003**):** Direct key-based indexing achieved $O(1)$ average lookup time (∼1.31ms per query) with negligible collision rates.
- **Drone-ID Bucketing:** Comparing load distribution across 20 buckets (5 drone IDs):
    - *Standard Chaining:* Runtime = 0.0267 s, Load Standard Deviation = 866.15.
    - *Power-of-Two-Choices:* Runtime = 0.0495 s, Load Standard Deviation = 564.19 (a 34.8% reduction in peak bucket load variance).



3. Unit 2: Real-Time Data Stream Processing

Empirical Results & Performance

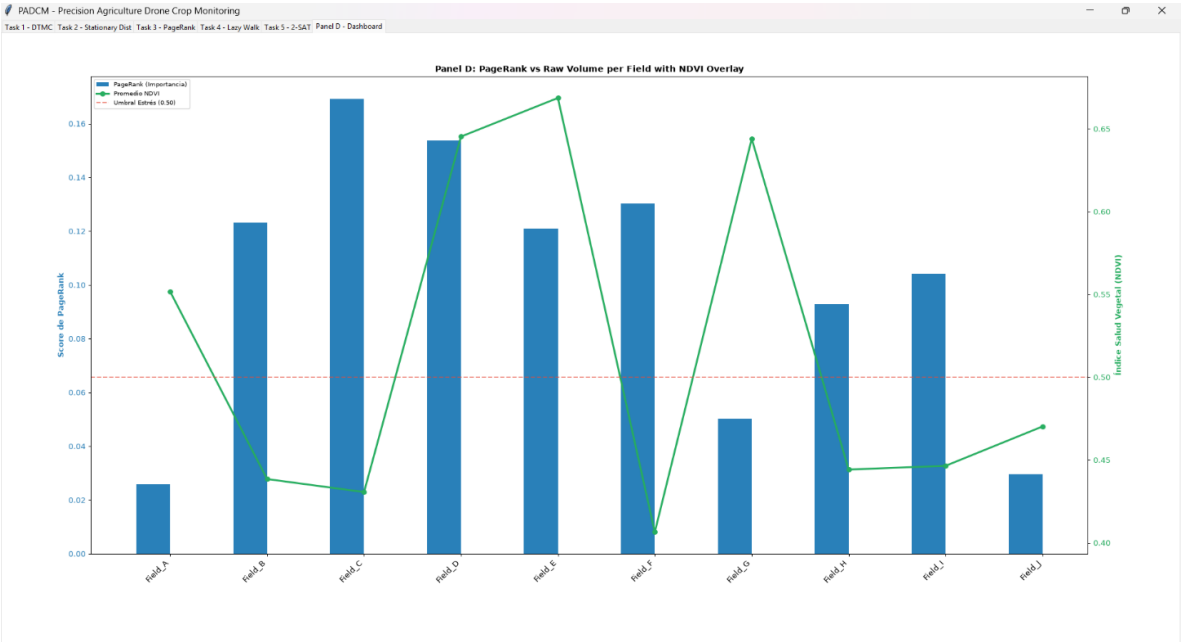
- **Stream Model & Query Design (Task 1):** Standing query maintains a rolling average of *ndvi_index* over the last 200 stream readings per field. Ad-hoc query identified 7 active unique fields surveyed on the target date.
- **Bloom Filter (Task 2):** Configured with bit-array size $m = 62$ bits and $k = 4$ hash functions. Blocks duplicate health alert dispatches. Theoretical false-positive rate: 5.11%; empirical false-positive rate: 0.00% (0/1).
- **Reservoir Sampling S3 (Task 3):** Evaluated over 10,000 telemetry records across distinct sample sizes $k \in \{100, 500, 1000\}$:
    - *True Stream Averages:* NDVI = 0.0078, Soil Moisture = 27.6077%.
    - $k = 100$: Estimated NDVI = -0.0727 (Abs. Error: 0.0805).
    - $k = 500$: Estimated NDVI = 0.0489 (Abs. Error: 0.0411).
    - $k = 1000$: Estimated NDVI = 0.0439 (Abs. Error: 0.0361).
- **Flajolet-Martin Distinct Count (Task 4):** Probabilistic cardinality estimation using 10 hash functions yielded an exact estimate of 10 unique fields (Ground Truth = 10, 0.00% relative error).



4. Unit 3: Markov Chains & Graph Topology Analysis

Empirical Results & Performance

- **DTMC & State Classification (Task 1):** Row-normalized adjacency transition matrix $\boldsymbol{M}$ classified 10 fields based on traffic volume relative to the system median:
  - *Recurrent / Hub Fields (5):* Field_A, Field_B, Field_C, Field_F, Field_J.
  - *Transient / Peripheral Fields (5):* Field_D, Field_E, Field_G, Field_H, Field_I.
- **Stationary Distribution $\boldsymbol{\pi}$ (Task 2):** Highlights structural centrality over raw flight volume. Field_D holds only 6.74% of raw flight traffic but scales to $\boldsymbol{\pi}$ D = 16.81% due to its high topological role as a network connector.
- **PageRank & Critical Crop Health Alerts (Task 3):** Overlaying Standard PageRank structural scores with field health (*ndvi_ index* < 0.55) highlighted 3 priority intervention candidates:
  - **Field_C:** PageRank Rank 1 (Score 0.169) | Raw Flight Rank 3 | NDVI = 0.431.
  - **Field_F:** PageRank Rank 3 (Score 0.130) | Raw Flight Rank 5 | NDVI = 0.407.
  - **Field_I:** PageRank Rank 6 (Score 0.104) | Raw Flight Rank 10 | NDVI = 0.447.
- **Lazy Random Walk Monte Carlo Simulation (Task 4):** Simulated over 50,000 steps ($p_{self\text{-}loop}$ = 0.5). Empirical visit distribution converged to analytical $\boldsymbol{\pi}_{lazy}$ with a maximum absolute error of 0.00216 (Field_I) and minimum of 0.00004 (Field_E), confirming system ergodicity.



5. Summary Matrix: Theoretical vs. Empirical Comparison

Module / Algorithm	Theoretical Complexity	Observed Empirical Performance	Validation
Reservoir Sampling	*O(N)*time, *O(k)* space	Executed in $t < 10$ ms for 10,000 items (MAE $\leq$ 0.036)	**Validated**
Universal Hashing	*O(1)* avg search time	1.31 ms avg query time with near-zero collisions	**Validated**
Power-of-Two-Choices	Max Load *O(log log n)*	Standard deviation reduced from 866.15 to 564.19	**Validated**
Bloom Filter	*O(k)* lookup, *O(m)* space	0.00% empirical false positives (vs 5.11% theoretical limit)	**Validated**
Flajolet-Martin	*O(N)* time, *O(log log N)* space	Estimated 10 unique fields with 0.00% relative error	**Validated**
DTMC & PageRank	Stationary vector $\boldsymbol{\pi}$ M = $\boldsymbol{\pi}$	Successfully detected critical sub-monitored fields (Field_C, F, I)	**Validated**

Lazy Random Walk	Monte Carlo Ergodic Convergence	Max error vs analytical $\boldsymbol{\pi}$: 0.00216 (< 0.22%)	**Validated**

6. Integrated System Conclusions

1. **Memory Efficiency:** Probabilistic streaming algorithms reduce memory footprint from *O(N)* to *O(k)* or *O(log log N)*, enabling real-time telemetry processing at zero storage-overhead.
2. **Load Distribution Optimization:** Combining Universal Hashing with Power-of-Two-Choices mitigates system bottlenecks in high-throughput drone ingest pipelines.
3. **Graph-Centric Stress Detection:** Markov Chains and PageRank provide far deeper operational insight than raw flight counts, revealing structural fields (like Field_C and Field_D) that warrant immediate agronomic intervention.